NATIONAL ENGINEERING COLLEGE, K.R.NAGAR, KOVILPATTI – 628 503

(An Autonomous Institution, Affiliated to Anna University Chennai)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

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DEPARTMENT OF INFORMATION TECHNOLOGY

INNOVATIVE TEACHING METHODOLOGY

Academic Year: 2026–2027 Odd Semester

COURSE CODE & COURSE NAME	**:**	**23CS31C/23IT33C – Computer Organization and Architecture**
TOPIC CONDUCTED ON	:	**Processor Organization and PC Assembly & Disassembly**

ACTIVITY NAME: Interactive Simulation and Hands-on Exploration of Processor Organization

Department	Date	Time	Venue
CSE	01.08.2026	11.10 am to 12.50 pm	ICL and AI&DL labs
IT	19.06.2026	11.10 am to 12.50 pm	Smart Classroom I

ACTIVITY DESCRIPTION:

A simulation-based practical learning activity was conducted for the Computer Organization and Architecture course to provide students with hands-on exposure to the internal working of a computer system. The activity integrated CompSimU-based processor simulation with PC assembly and disassembly to help students relate theoretical concepts with practical system operations.

Using the CompSimU simulation environment, students explored instruction execution, register operations, processor organization, and the interaction of various CPU components. The simulation enabled students to observe the execution of instructions step by step and understand how data moves within the processor.

As part of the hardware-oriented activity, students participated in PC assembly and disassembly, where they identified and examined major components such as the motherboard, processor, RAM, SMPS, and storage devices. Students also observed the physical connections between the components and verified the proper arrangement of hardware during assembly.

The activity encouraged students to observe, analyze, discuss, and relate the simulated processor operations with the actual hardware components. Thus, the activity provided an integrated learning experience by connecting COA concepts, processor-level operations, and computer hardware architecture.

ACTIVITY OBJECTIVE:

The activity was conducted with the following objectives:

- To provide students with practical exposure to fundamental Computer Organization and Architecture concepts.
- To help students understand instruction execution, register operations, and processor organization through simulation.
- To familiarize students with the internal hardware components of a computer system through PC assembly and disassembly.
- To enable students to relate theoretical processor concepts with real-world hardware operations.
- To develop students' observation, analytical, troubleshooting, and collaborative learning skills.

ABOUT SIMULATION TOOL:

CompSimU (Computer Simulator Universal) is an educational simulation tool used to understand the internal operations of a computer system. It enables students to visualize the execution of instructions within the processor and observe the interaction among different hardware components during program execution.

The simulation activity focused on:

- Instruction execution
- Register operations
- Processor organization
- Data movement within the processor
- Understanding CPU components
- Step-by-step observation of processor operations

ACTIVITY IMPLEMENTATION:

The activity was carried out in the following stages:

Stage 1 – Introduction to Processor Organization

Students were introduced to the major components of a processor and their roles in executing instructions. The relationship between the CPU, registers, memory, and other system components was discussed.

Stage 2 – CompSimU Installation and Demonstration

The CompSimU software was installed and demonstrated to the students. Students were introduced to the simulation environment and its features for observing processor operations.

Stage 3 – Instruction Execution Simulation

Students observed the execution of instructions through the simulator. The step-by-step execution helped them understand how instructions are processed within the CPU and how registers participate during execution.

Stage 4 – PC Disassembly

Students examined an actual computer system by identifying its major internal components. They observed the motherboard, processor, RAM, SMPS, and storage devices and studied their roles in the overall system.

Stage 5 – PC Assembly

Students participated in assembling the computer system and verifying the connections between its components. This provided practical exposure to computer hardware organization and interfacing.

Stage 6 – Simulation and Hardware Correlation

Students compared the processor operations observed through CompSimU with the physical components examined during PC assembly and disassembly. This helped them establish a connection between theoretical COA concepts, simulated operations, and real computer hardware.

STUDENT ENGAGEMENT:

The activity emphasized active student participation rather than conventional classroom instruction.

Students were encouraged to:

- Observe processor operations through simulation.
- Identify and examine computer hardware components.
- Participate in PC assembly and disassembly.
- Discuss the function of individual components.
- Relate simulated instruction execution to actual processor operations.
- Verify hardware connections during assembly.
- Share their observations and learning with peers.

The practical approach encouraged participation and discussion among students.

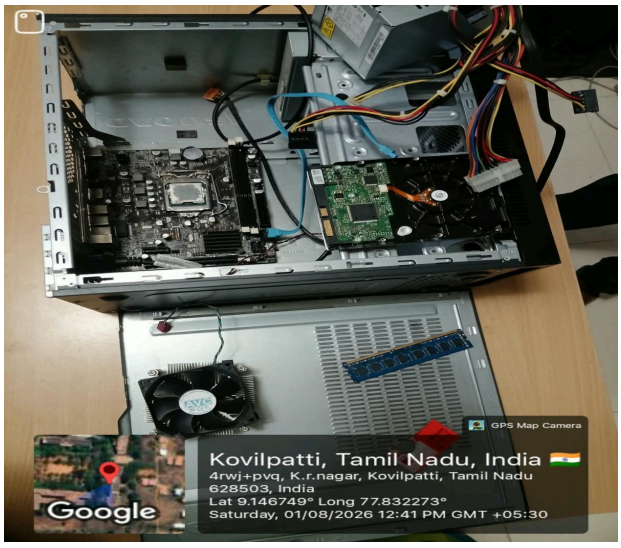
ACTIVITY OUTCOMES:

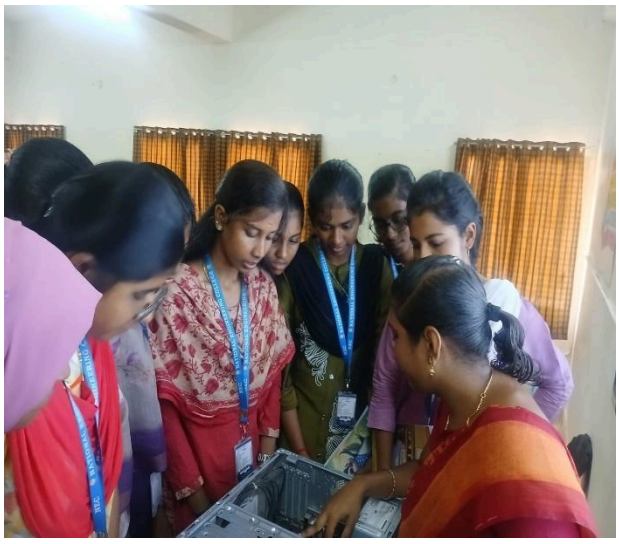
- Students gained practical understanding of computer organization concepts beyond theoretical learning.
- Students understood the step-by-step execution of instructions within the processor using simulation.
- Students identified and understood the functions of major physical components of a computer through PC assembly and disassembly.
- Students effectively connected processor operations observed in simulation with actual hardware components.
- Students developed practical observation, analytical, and troubleshooting skills through hands-on activities.
- The activity encouraged active participation, peer learning, and technical discussion among students.

ACTIVITY SNAPSHOTS:













Content Available at:

https://drive.google.com/drive/folders/17tDWUSDaRg0j5PAEAuuW3u7TZ5zdm1xP?usp=sharing

Course Instructors:

1. Dr.J.Ida Christy, AP(SG)/CSE
2. Ms.N.Gowthami, AP(SG)/IT
3. Ms.R.Navedha Evanjalin, AP/CSE
4. Ms.M.Vaijayanthimala, AP/CSE

FEEDBACK OF STUDENTS

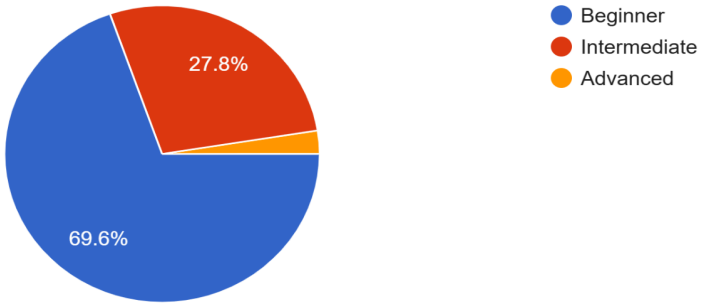
ACTIVITY LEARNING: Interactive Simulation and Hands-on Exploration of Processor Organization

[Response collected through Google Forms]

COA TLP Responses

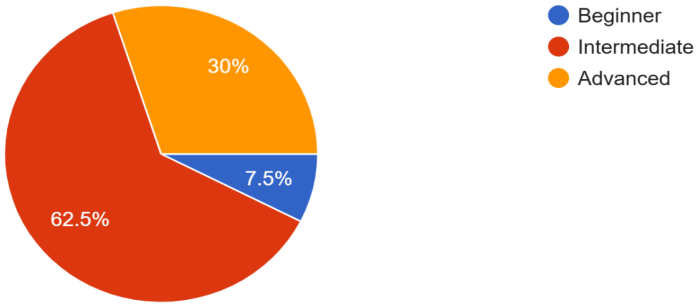
How would you rate your knowledge of PC assembly before the workshop?

79 responses



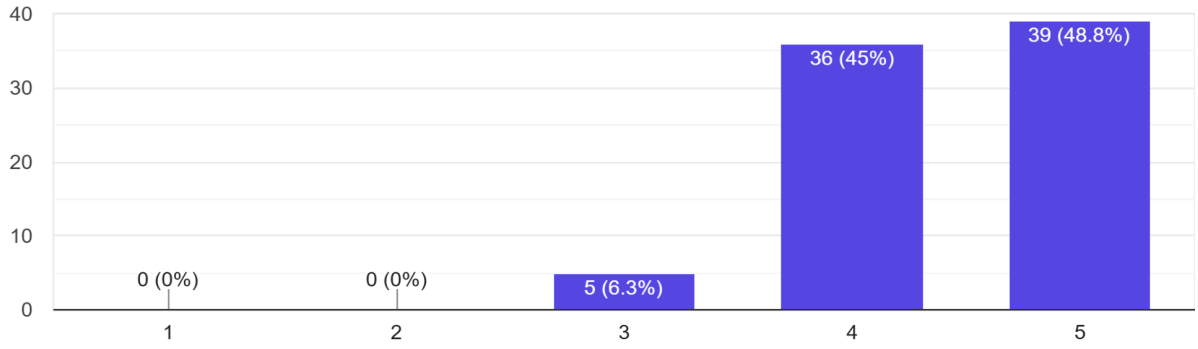
How would you rate your knowledge of PC assembly after the workshop?

80 responses



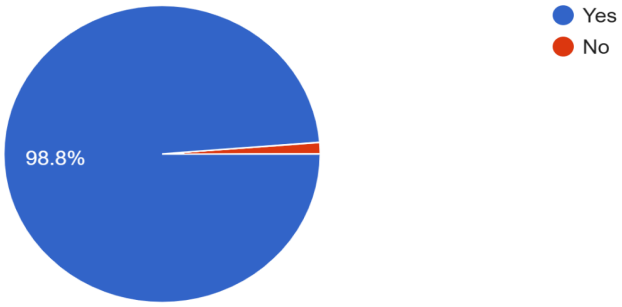
How well did you understand the PC components after the session?

80 responses



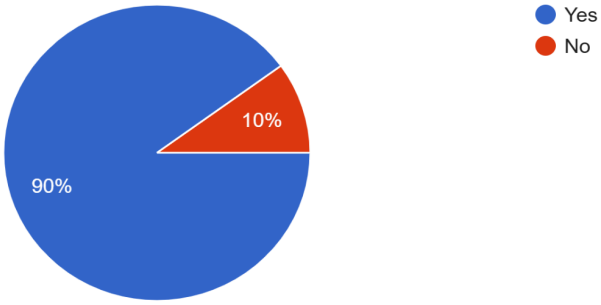
Did the practical session improve your knowledge?

80 responses



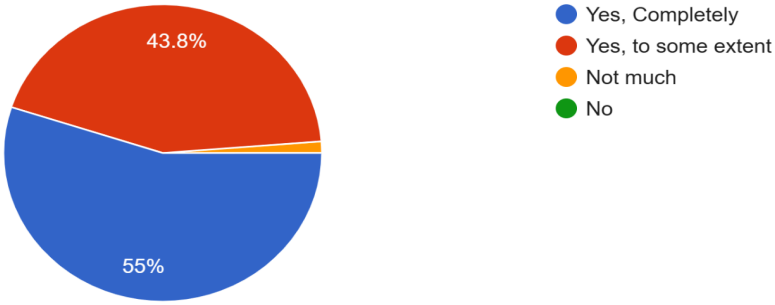
Can you install the motherboard correctly

80 responses



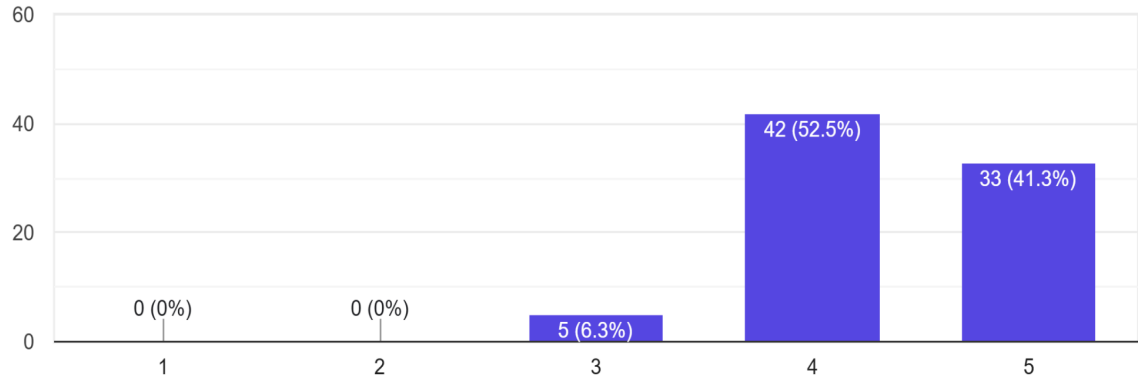
Did you learn how to identify computer hardware components?

80 responses



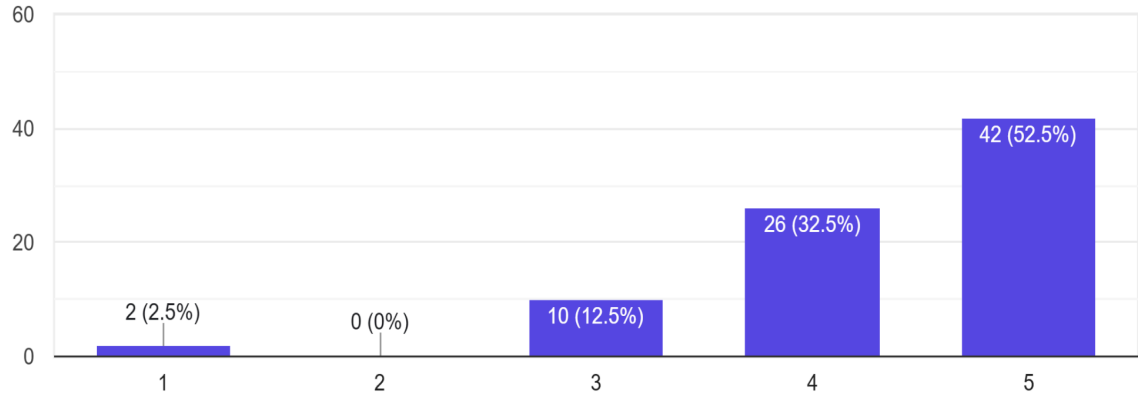
How well did you understand the correct sequence of PC assembly?

80 responses



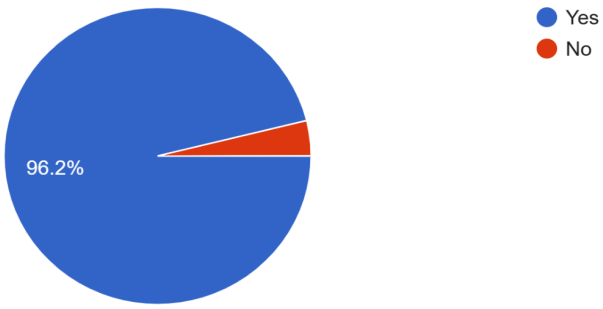
How easy was it to use COMPSIMU?

80 responses



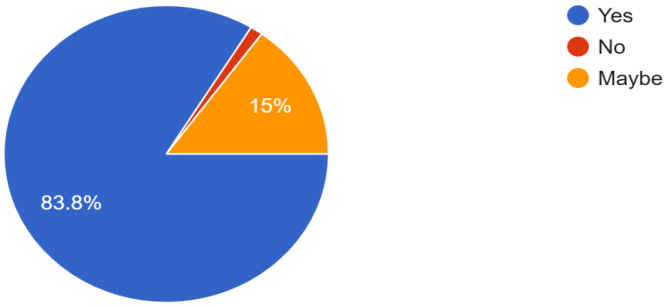
Did COMPSIMU help you understand the PC assembly process better?

78 responses



Would you like to use COMPSIMU for future hardware learning sessions?

80 responses



How did the Simulation and Analysis Integrated Learning activity enhance your understanding of Computer Organization and Architecture concepts?
The activity helped me understand processor organization and instruction execution more clearly through practical simulation.
The CompSimU simulation helped me visualize the step-by-step execution of instructions and relate it to the concepts discussed in class.
The PC assembly and disassembly activity improved my understanding of the functions of different computer hardware components.
The activity helped me connect theoretical COA concepts with their practical implementation in a computer system.
The combination of simulation and hands-on hardware activities made the learning process more interactive and improved my practical understanding.

How did the activity on Processor Organization and PC Assembly & Disassembly help you understand the concepts of Computer Organization and Architecture better?
The activity helped me understand how the different components of a computer system work together instead of learning their functions only through theory.
By studying and assembling the components of a PC, I was able to clearly understand the role of the processor, memory, motherboard, storage devices, and other peripheral components.
The hands-on experience helped me connect theoretical concepts of processor organization with the actual hardware components present inside a computer system.

The PC assembly and disassembly activity improved my practical knowledge and helped me understand how different hardware components are connected and communicate with each other.
The activity made the concepts of Computer Organization and Architecture more interactive and meaningful, giving me confidence in identifying and understanding the functions of various computer components.

1) 8/9/26
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4) 08/09/26

COURSE INSTRUCTORS

8/9/26

COURSE CO-ORDINATOR

8/9/26
HOD/CSE

8.9.26
HOD/IT

DEAN ACADEMIC

PRINCIPAL